



[Tech Talk] Next.js - Jake

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≡ Description	Explains how Next.js works and how it can fit into our future projects.
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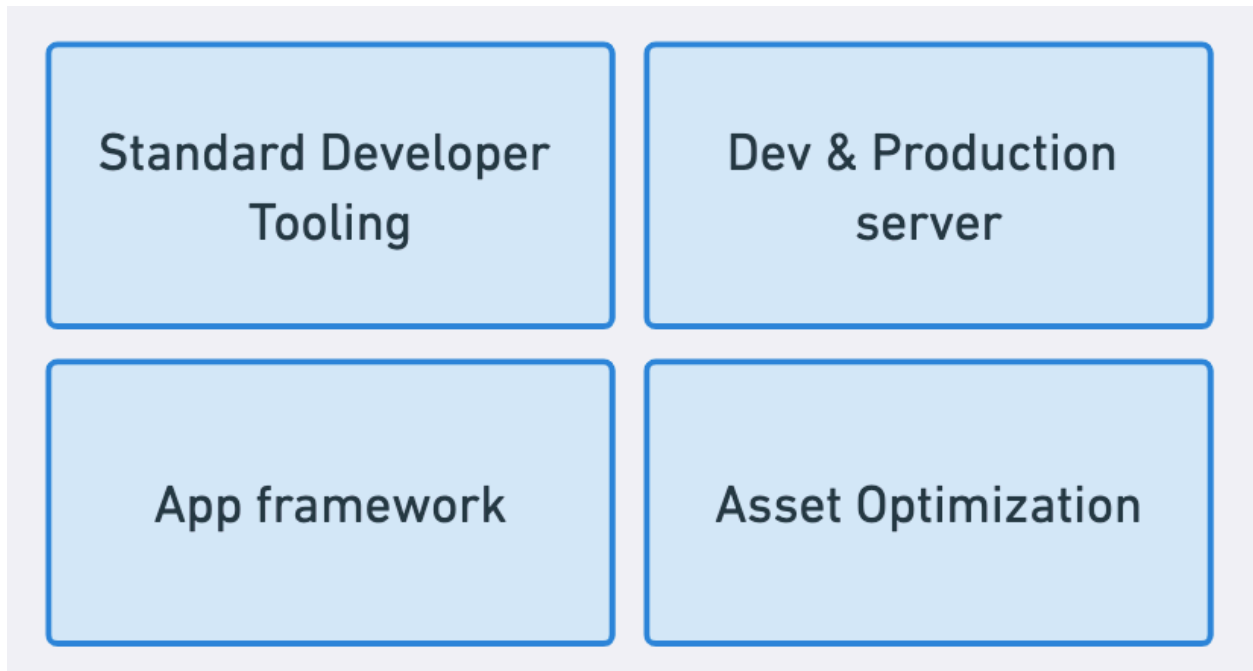
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What is Next.js?

- "The react framework for the web"
- Built and open-sourced by Vercel

Where it fits



- Replacement for Create React App

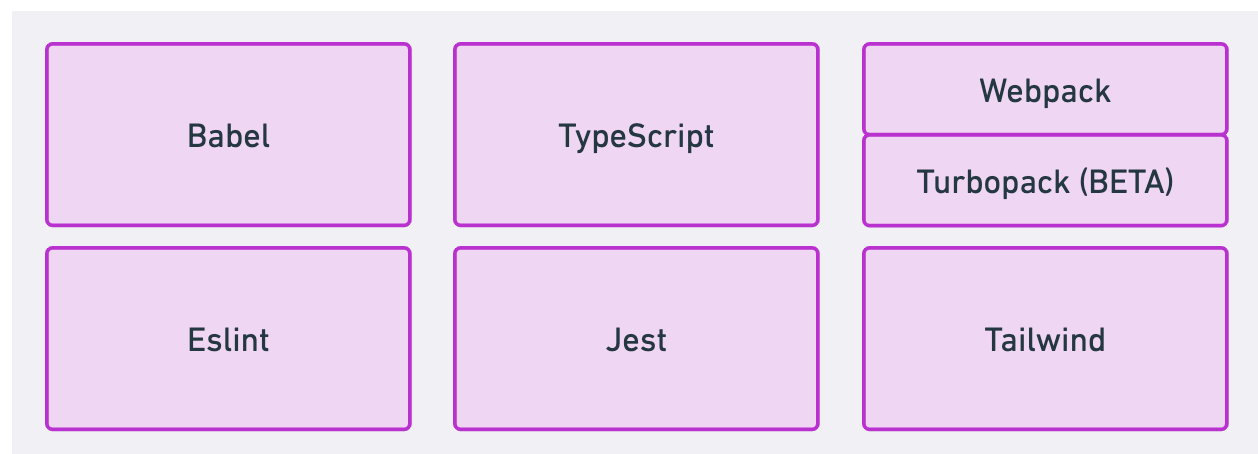
Currently, we use CRA to set up our app's tooling. We have a couple of packages on top of CRA like Craco for config overrides, and this would replace all of that.

- A framework of tools to make web development quick and painless.

NextJS would live in the client *code* but it would also be a server runtime on the backend. This runtime doesn't require development or configuration aside from deployment.

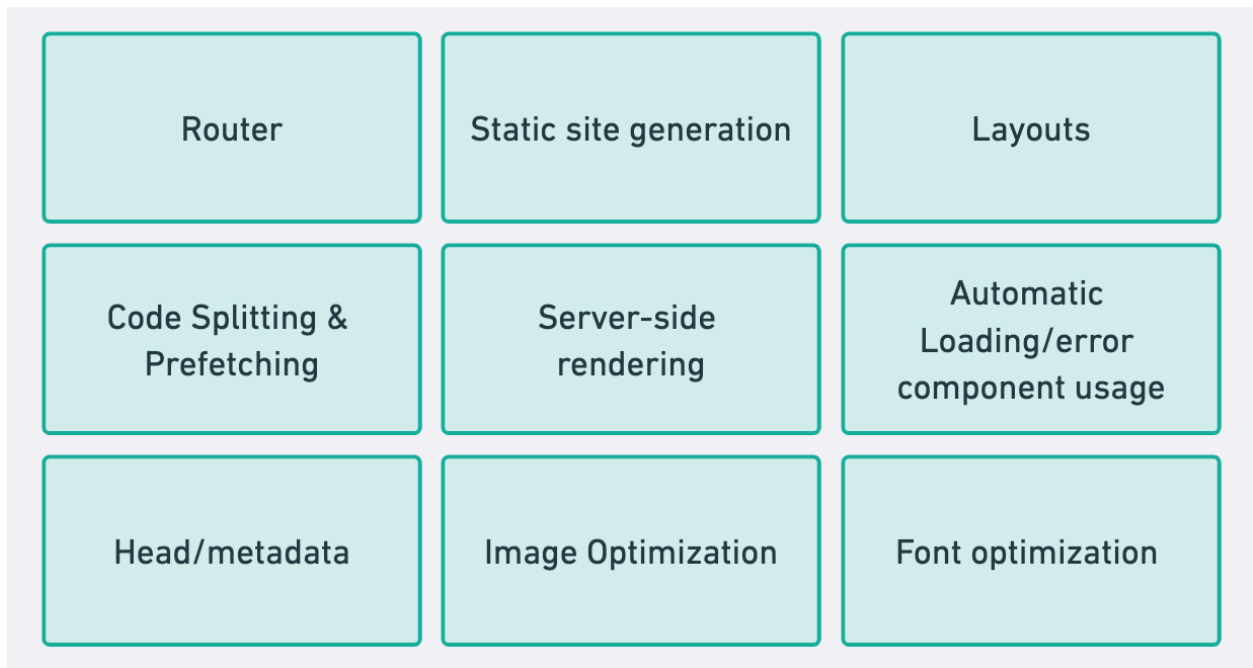
How Next.js works

Tooling

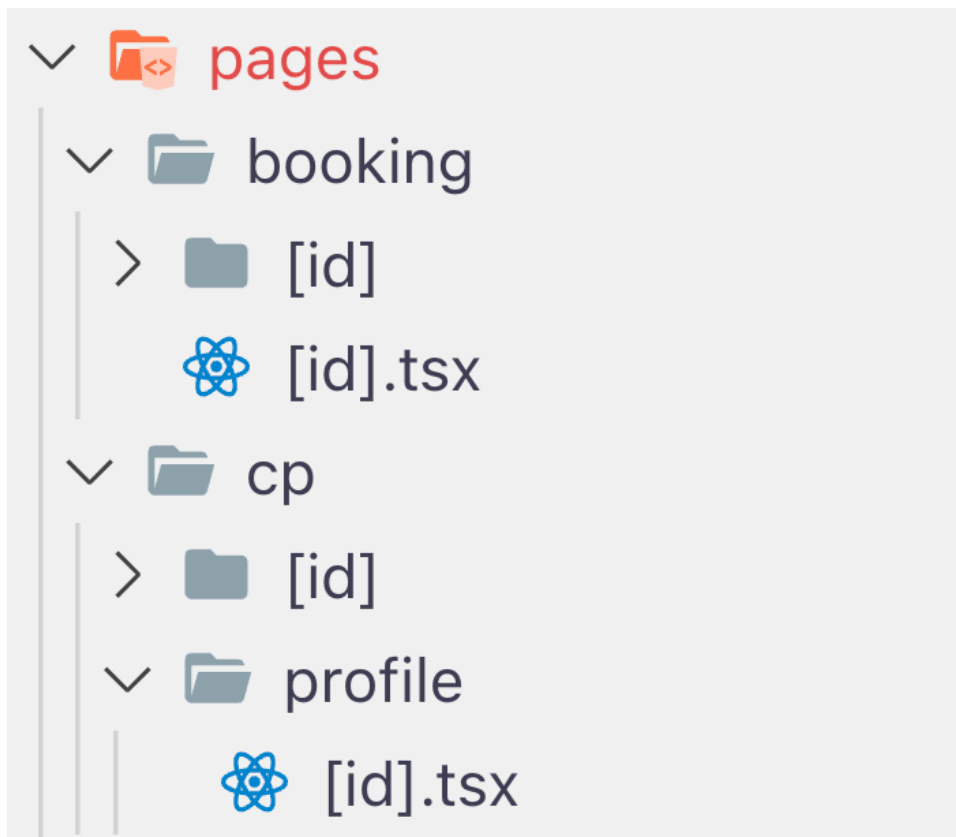


- Manual overrides and escape hatches are easy to find and follow conventions

Application Framework



Routing



- Filepath-based routing makes the correct page code easy to find and enforces conventions on page names and URL patterns
- Instead of many separate exports, like react-router-dom (i.e. `useNavigation`, `useParams`, `useSearchParams`), the router is accessible through a single `useRouter` hook.
- Automatically supports code splitting

Route (pages)	Size	First Load JS
o /	1.54 kB	342 kB
o /_app	0 B	269 kB
o /404	182 B	269 kB
o /booking/[id]	5.97 kB	403 kB
o /booking/[id]/report	10.6 kB	376 kB
o /booking/[id]/review	8.88 kB	337 kB
o /cp/[id]/rebook	9.29 kB	337 kB
o /cp/profile/[id]	7.27 kB	335 kB
o /forward/[hash]	569 B	270 kB
o /help	1.33 kB	271 kB
o /login	2.89 kB	300 kB
o /messages	3.44 kB	332 kB
o /messages/[channelid]	8.77 kB	345 kB
o /new	1.39 kB	271 kB
o /oldIndex	227 B	269 kB
o /past	2.67 kB	341 kB
o /settings	893 B	270 kB
o /settings/account	3.52 kB	300 kB
o /settings/Address	3.58 kB	300 kB
o /settings/cleaning_notes	5.17 kB	302 kB
o /settings/credit_card	7.59 kB	304 kB
o /settings/membership	10.3 kB	338 kB
o /settings/recurring_frequency	3.47 kB	368 kB
o /upcoming	1.76 kB	342 kB
+ First Load JS shared by all	269 kB	
chunks/framework-2c79e2a64abdb08b.js	45.2 kB	
chunks/main-6e75805f805dee0.js	33.2 kB	
chunks/pages/_app-ed62faa4d3cafff0a.js	190 kB	
chunks/webpack-2f903acb0cccbf9e.js	1.15 kB	
css/9fd33e749c746ee3.css	137 B	

Runtime

- Dev server with hot module reloading
- Production typically run with a persistent nodejs server
 - Prerendering

Deployment

- Generally requires BE server running

- **Can** be exported statically, but I'd need to fully test that to make sure it all works. This wouldn't work for some optimizations like SSR or potentially code splitting.
- Vercel
 - Supports PR deploy previews (already done)
 - Built-in analytics
 - near-zero config for Next.js
 - DDoS mitigation
 - Edge execution for SSR
 - CDN asset caching
 - Supports SWR
 - Additional Links

Platform Limits
- Cloudflare, AWS Amplify also good options for hosting.
 - [AWS thread on reddit](#)

The Future

The NextJS conf's keynote is a great video to watch for understanding the longer-term goals and ambitions for the framework. [Video here](#).

Turbopack

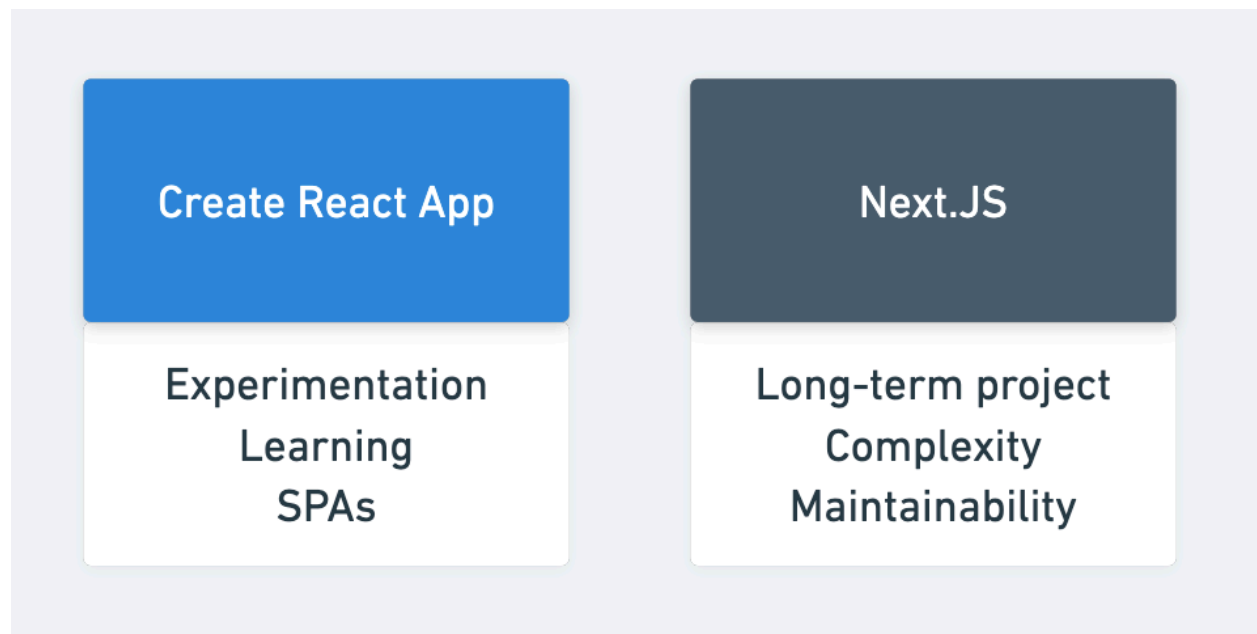
Next.js 13 supports a beta `/app` directory to eventually replace the current `/pages` directory.

- **app Directory (beta):** Easier, faster, less client JS.
 - **Layouts**
 - **React Server Components**
 - **Streaming**
- **Turbopack (alpha):** Up to 700x faster Rust-based Webpack replacement.
- **New next/image:** Faster with native browser lazy loading.
- **New @next/font (beta):** Automatic self-hosted fonts with zero layout shift.
- **Improved next/link:** Simplified API with automatic `<a>`.

Summary of Next.js v13 from their release [blog post](#)

Differences Between Next/CRA

Project Goals

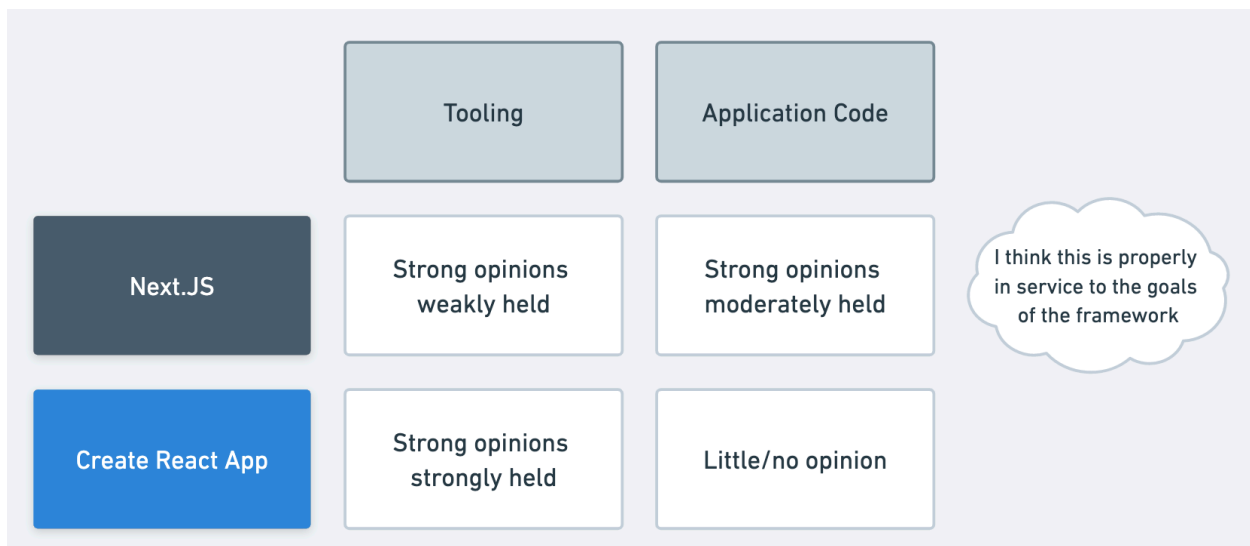


CRA, because of how it does tooling, is aimed at **SPAs and newer development**. It is a great sandbox to spin up small apps and experiments but once things get complicated, it's hard to work with due to its configuration rigidity.

- For these reasons, CRA doesn't support server-side rendering or great code splitting out of the box.

Next is built for larger websites and apps, with a server-first roadmap to support dynamic content with a good UX and DX. It sets up repeatable structures that are easy to find documentation and examples for.

Opinions & Framework Decisions



Both CRA and Next are opinionated. That's why they exist—to make choices for developers to speed up project setup and development.

CRA is **decisive** about **what** tools are used **and how** those tools are used. It has little to no preference for how applications are structured.

Next.js is **opinionated** about **what** tools are used, less about **how** they are used. It holds more opinions on how the application code is written, like recommending directory names and encouraging usage of Next's asset optimization components.

- This could be seen as a downside, but with the rapid development of FE capabilities, having a tool help keep projects unblocked from evolving to support future React features and/or browser features is really nice.

Future & Project Maintenance

- Vercel has a profit motive for keeping Next.js on the cutting edge and easy to use.
- Create react app hasn't had a single commit to the project since September 2022.
 - Used to be recommended on react's docs, but that recommendation has been dropped in place of Create next app, a CLI to spin up a Next.js project

Should we use Next.js?

It's not a zero-effort or zero-cost move, but the goals of Next more closely align with what we're trying to build.

Performance isn't a top priority right now, but if we wait too long, it could become an unpleasant bottleneck.

Since CRA isn't maintained anymore, and won't track updates to the tools it uses, continuing to use it will only end up being technical debt later on.

Regarding risk, since Next.js is a framework for React and uses common tools, there will always be a migration path away from it. Vercel obviously has an interest in encouraging the use of their deployment services, but they do not enforce any such requirement.

Rough Plan & Ideas

I've already been working on moving the current New C Web project to Next.js in my spare time. It's not a small change, and there's a risk involved with large changes and introducing regressions.

I think while we're at a reduced traffic load to New C Web, now is the time to consider this move.

Vercel isn't a **cheap** hosting service per se, but I think we should evaluate the benefits of using Vercel.